A RING MAP IS PERFECT EXACTLY WHEN THE TARGET HAS A FINITE RESOLUTION BY FINITE PROJECTIVES OVER A POLYNOMIAL ALGEBRA

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot; nothing was added, removed or redefined and no package was added.

Editorial note. The source URL is the source-level Stacks Project URL, as required when more than one card is produced from one source; the per-claim tag is carried by claim_key.

Editorial note. The author writes 'a $A[x_1, \ldots, x_n]$-module' in the proof where 'an' would be expected; that is the author's text and is reproduced unchanged. The two references inside the counted proof span are recorded as non-load-bearing navigation pointers.

Allowed standard prerequisites (editorial).

- Pseudo-coherence of a ring map checked after a polynomial presentation. used at proof lines 21782 and 21791-21792. Is one of the two ingredients in the definition of a perfect ring map.
- Finite tor dimension of the target over the base. used at proof lines 21783 and 21793-21794. Is the other ingredient, supplied in the converse direction by flatness of the resolution.
- Perfectness of a module over a polynomial ring and its equivalence with the existence of a finite resolution by finite projectives. invoked at proof lines 21784-21787. Converts the perfectness assumptions into the resolution statement and back.

Source reference. Original lemma 18917: sources/stacks-more-algebra/pinned-original-source.tex, source lines 18917–18917. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Source reference. Original lemma 20175: sources/stacks-more-algebra/pinned-original-source.tex, source lines 20175–20175. Same-chapter navigation pointer; the statement it names is standard for the section and the argument that consumes it stays inside the counted proof span.

Lemma 0.2. *A ring map $A \to B$ is perfect if and only if $B = A[x_1, \ldots, x_n]/I$ and B as an $A[x_1, \ldots, x_n]$-module has a finite resolution by finite projective $A[x_1, \ldots, x_n]$-modules.*

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Proof. If $A \to B$ is perfect, then $B = A[x_1, \ldots, x_n]/I$ and B is pseudo-coherent as an $A[x_1, \ldots, x_n]$-module and has finite tor dimension as an A-module. Hence Lemma 20175 implies that B is perfect as a $A[x_1, \ldots, x_n]$-module, i.e., it has a finite resolution by finite projective $A[x_1, \ldots, x_n]$-modules (Lemma 18917). Conversely, if $B = A[x_1, \ldots, x_n]/I$ and B as an $A[x_1, \ldots, x_n]$-module has a finite resolution by finite projective $A[x_1, \ldots, x_n]$-modules then B is pseudo-coherent as an $A[x_1, \ldots, x_n]$-module, hence $A \to B$ is pseudo-coherent. Moreover, the given resolution over $A[x_1, \ldots, x_n]$ is a finite resolution by flat A-modules and hence B has finite tor dimension as an A-module. $\square$